Learning Curves

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 27: compare results under shared evaluation conditions.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Read training-size learning curves. Try the recall before opening notes.

Your learning target

I can read **training**-size **learning curve**s.

I can describe **variability** across **repeat**ed development runs.

I can avoid unsupported guarantees from a trend.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

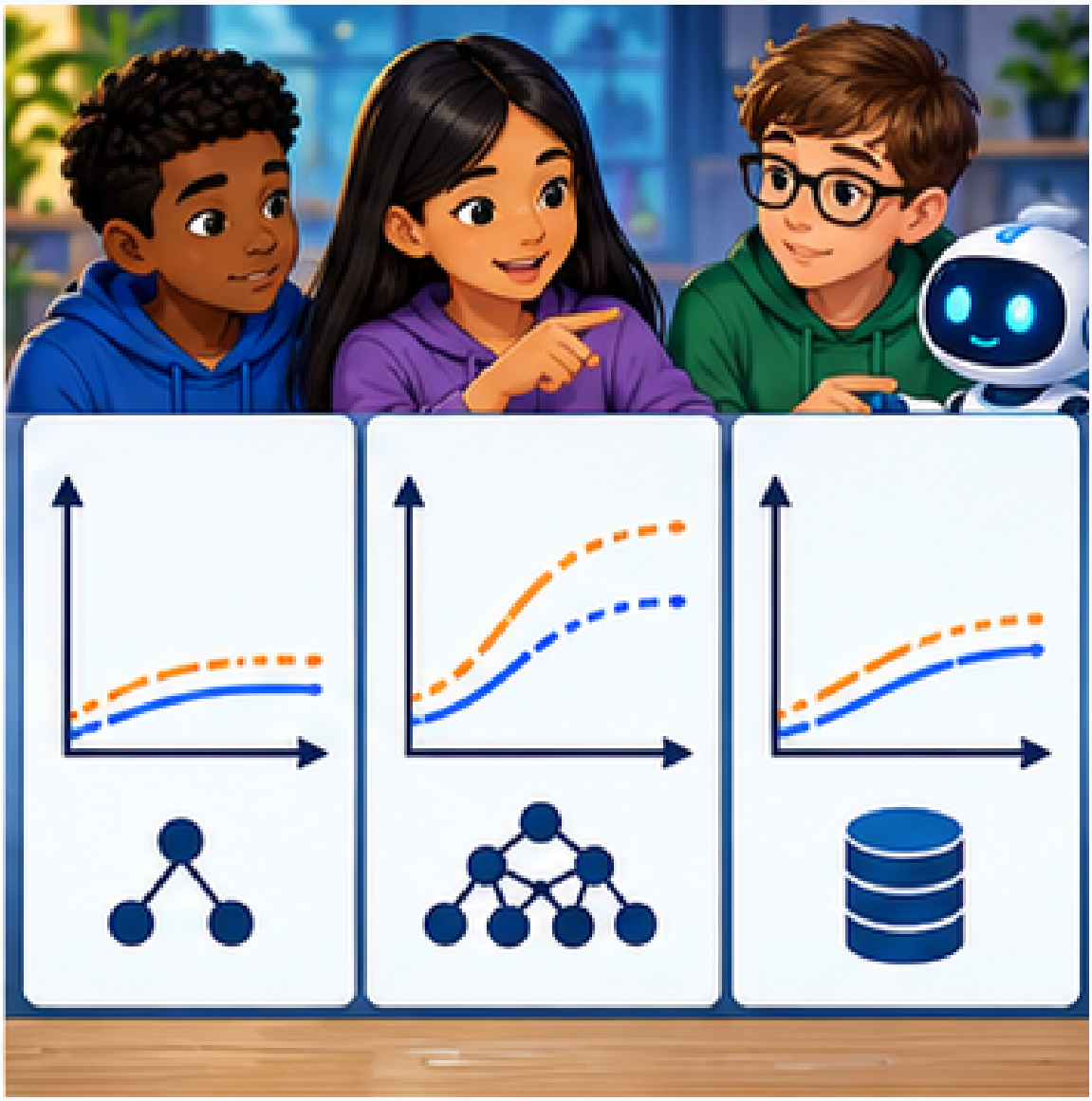
Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Learning curves plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.



Your target

LEARNING TARGET **Training** sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

Builder name and experiment date:

Readiness for the connected explanation

What does validation loss add to **training** loss?

This week's end goal

By the end, I can explain **learning curve**s using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

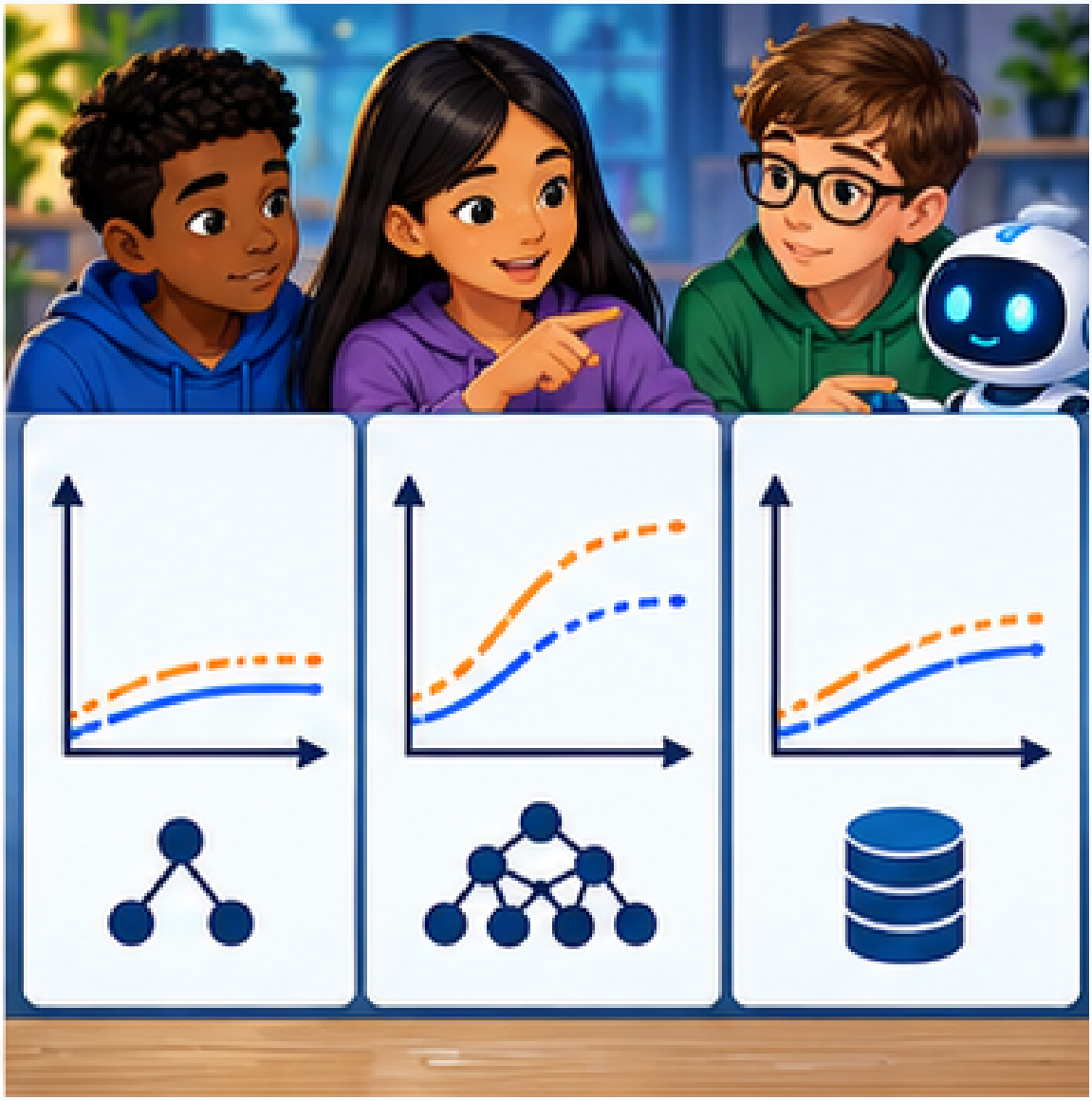
Notice the evidence

> Investigate the supplied Learning Curves case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **learning curve**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A **learning curve** asks how performance changes as the amount of **training evidence** grows. Keep the validation protocol, model settings and metric comparable.

Use subsets of **training** only; never add validation rows into the **training** subsets. Multiple **repeat**s reveal sensitivity to sample selection or model randomness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **learning curve**s and **variability**.

STUDENT LESSON · 3 OF 18

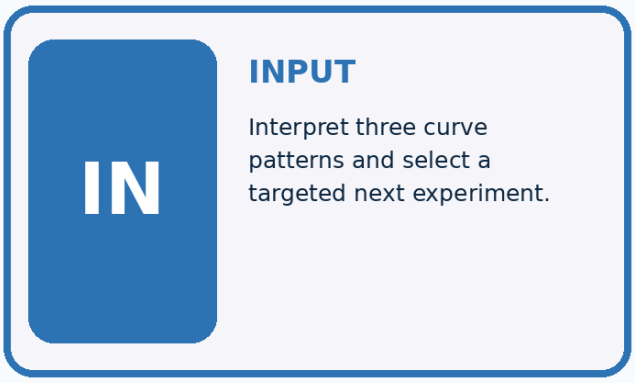
See how it works

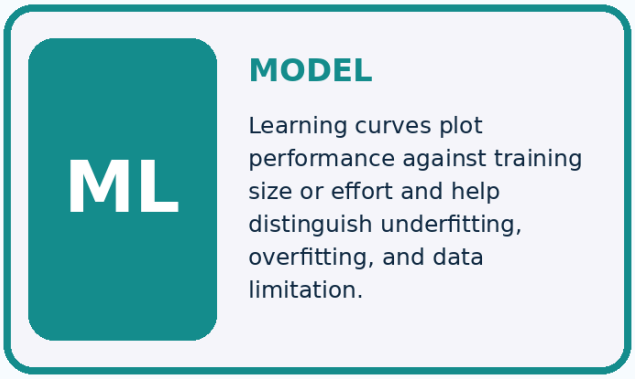
> Follow the mechanism for Learning Curves. Connect each labelled input, deciding step and result.

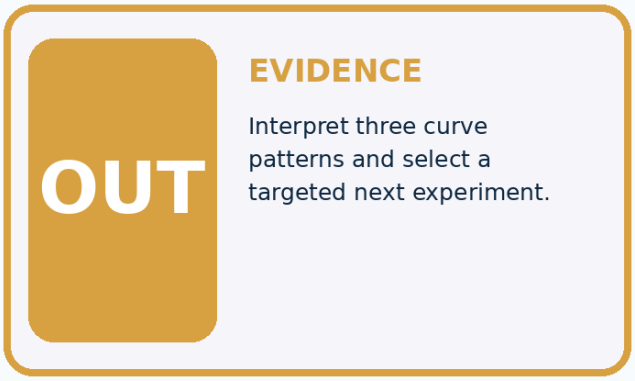
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Interpret three curve patterns and select a targeted next experiment.

MODEL IDEA **Learning curve**s plot performance against **training** size or effort and help distinguish under**fitting**, over**fitting**, and data limitation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Both train and validation scores are low and close: likely under**fitting**. Train high, validation much lower: likely high **variance** or over**fitting**.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

At **training** sizes 10, 20, 40, validation MAEs across three declared **repeat**s are [7, 9, 8], [5, 7, 6], [4, 6, 5]. Means are 8, 6, 5 and **range**s 2, 2, 2. Mean **training** MAEs are 1, 2, 3.

Validation improves as **training** grows, while **training** error rises because **fitting** more varied examples can be harder.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

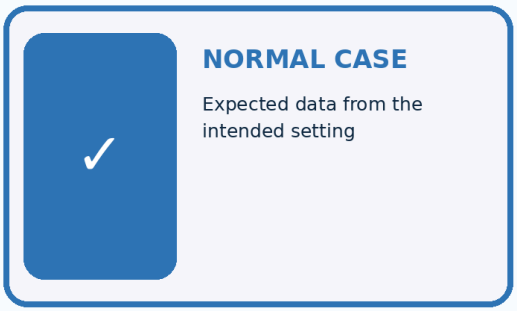
Compare the cases

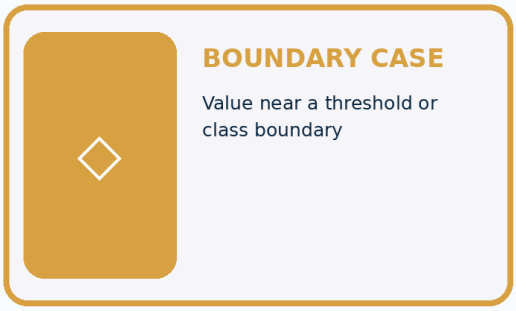
> Compare the supplied cases for Learning Curves. Identify the detail that changes the conclusion.

EVALUATION DESIGN

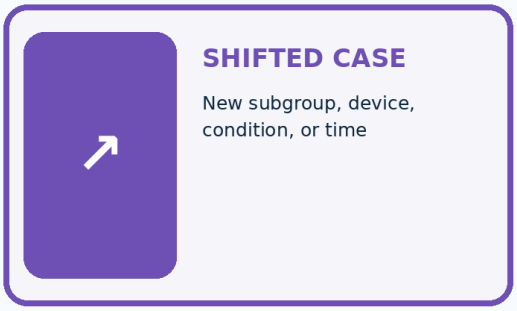
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: A30-plant experiment gives errors [4, 4, 4].

Predicted result

Prediction: Mean remains4; this additional size does not improve the observed mean over20 plants.

Why the result follows

Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Learning Curves to explain a concrete decision, rather than only memorising definitions.

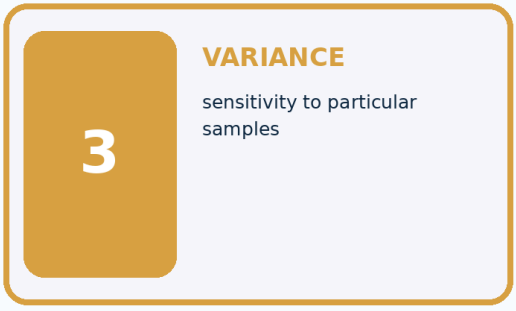
VOCABULARY

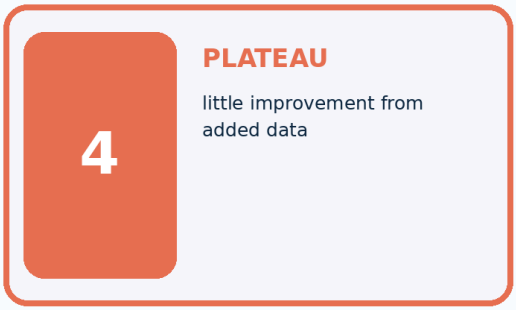
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
learning curve	performance across training sizes	
bias	systematic underfitting	
variance	sensitivity to particular samples	
plateau	little improvement from added data	

MISCONCEPTION TO REPAIR Every **learning curve** that rises proves more data will solve the problem.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Learning curve	Performance plotted against training sample size
Repeat	Another run under a recorded procedure
Variability	How much results differ between runs
Range	Maximum minus minimum of observed values

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Learning Curves, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 28 • WORKED EVIDENCE

1. Both train and validation scores are low and close: likely underfitting.
2. Train high, validation much lower: likely high variance or overfitting.
3. Validation improves as data grows and gap narrows: more representative data may help.
4. Curves suggest hypotheses; confirm with controlled experiments and error slices.

Step 1

STEP 1 Both train and validation scores are low and close: likely under**fitting**.

Step 2

STEP 2 Train high, validation much lower: likely high **variance** or over**fitting**.

Step 3

STEP 3 Validation improves as data grows and gap narrows: more representative data may help.

Step 4

STEP 4 Curves suggest hypotheses; confirm with controlled experiments and error slices.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Learning curves and variability at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
10 training rows	Validation 7 9 8	Mean 8 range 2
20 training rows	Validation 5 7 6	Mean 6 range 2
40 training rows	Validation 4 6 5	Mean 5 range 2

The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Why can training error rise while validation error falls? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learning Curves

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

The facts we will use

Supplied facts

At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/28

Learning Curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Summarise first size

Mean6 errors; range5-7, corresponding to mean 40% accuracy.



2. Summarise larger size

Mean4 errors; range3-5, corresponding to mean 60% accuracy.



3. State bounded pattern

More training plants improved the average in these supplied runs; variability remains.

CHANGE ONE THING

A30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Summarise first size	Mean6 errors; range5–7, corresponding to mean 40% accuracy.
2. Summarise larger size	Mean4 errors; range3–5, corresponding to mean 60% accuracy.
3. State bounded pattern	More training plants improved the average in these supplied runs; variability remains.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

Keep the stated limit

Read the labelled picture: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A30-plant experiment gives errors [4, 4, 4]. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

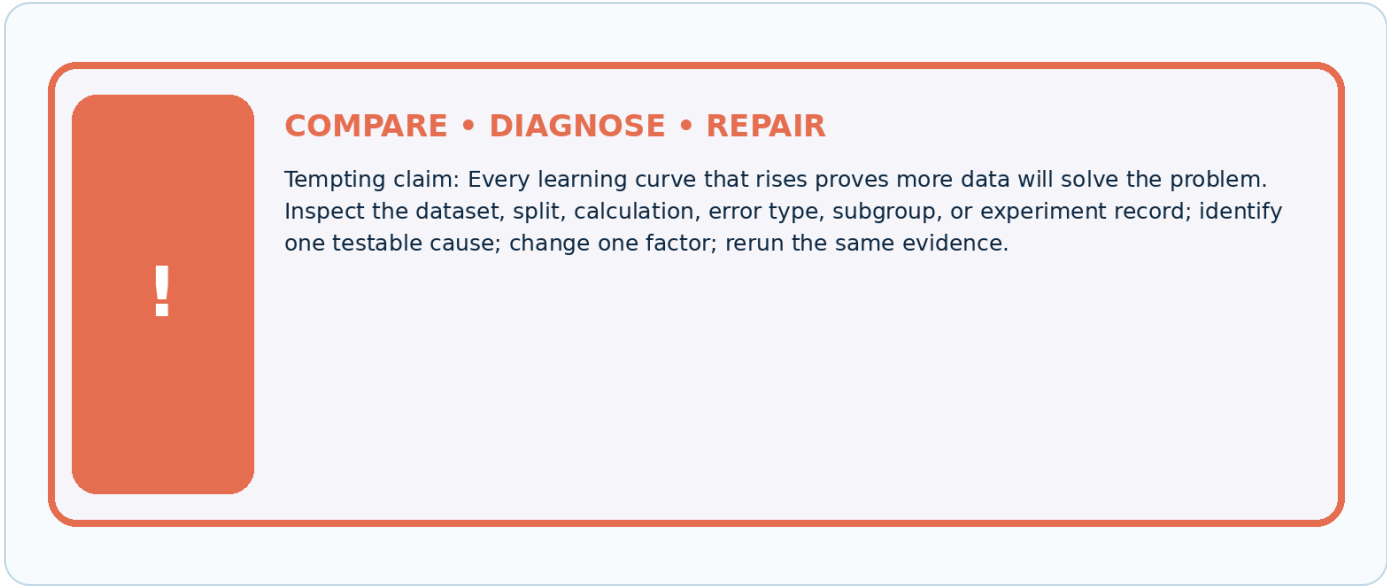
> Find the unsupported step in the Learning Curves claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

> CLAIM TO CHALLENGE Every **learning curve** that rises proves more data will solve the problem.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Collect additional data because it addresses a known coverage need, not merely to enlarge a count. When the curve stops improving, inspect labels, features and task suitability.

Keep the final test reserved while planning these changes.

Why is an observed range not a confidence interval?

STUDENT LESSON · 8 OF 18

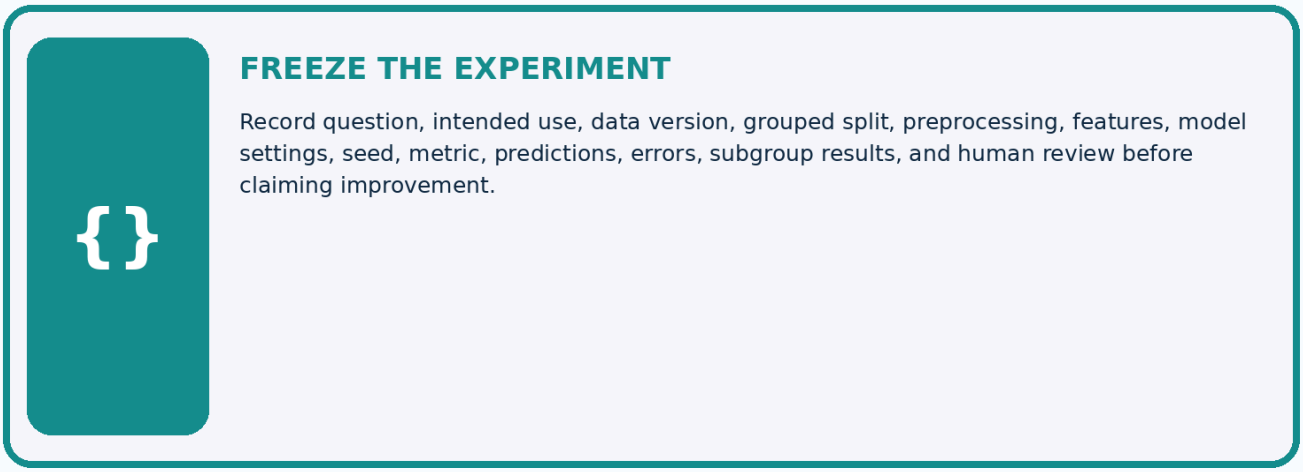
Practise together

Practise Learning Curves with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Record the actual subset identifiers and seeds for every **repeat**.

If the validation population changes at the same time as **training** size, label that as a different experiment rather than attributing the difference only to more **training** data.

Practise with less help: Read both curves

Use workbook W2 for the connected example "**Learning curve**s and **variability**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Learning curve**s and **variability**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Learning Curves. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

1. Write the hypothesis and expected **evidence** before running.

2. Curve Clinic: match three graphs to diagnoses, challenge each diagnosis, and propose the smallest informative experiment.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

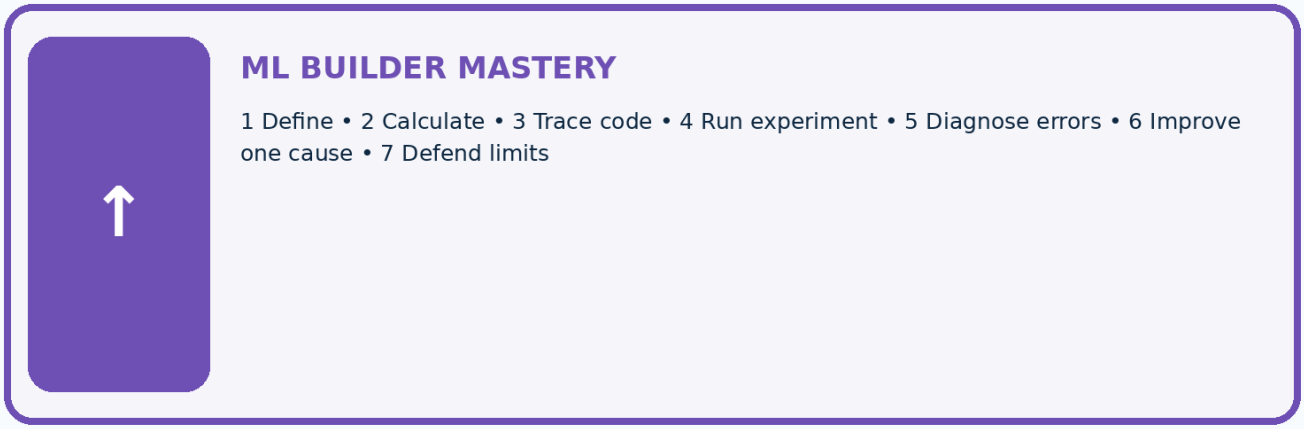
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Learning Curves. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **learning curve**, **bias**, **variance**, **plateau**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Every **learning curve** that rises proves more data will solve the problem.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Plot Plant Classifier train/validation performance at several training sizes and document the chosen next action.

THIS WEEK'S CHECKPOINT Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Interpret three curve patterns and select a targeted next experiment.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Feature Engineering.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Learning Curves case for supported practice. Repeated examples do not become fresh assessments.

Week 28 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

The question

Interpret the curve and uncertainty.

Worked explanation

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

More training data on the same validation set

Evidence or stage	Value or result
20	60
40	70
80	78

With20,40,80 **training** leaves, fixed-set validation accuracy is**60%**,**70%**,78%.

Interpret the curve and uncertainty.

These runs improve with more data. Further representative data may help, but gains are not guaranteed. Controlled **repeat**s reveal **variability**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/**range**s and a careful comparison.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 28 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> As **training** examples grow from 50 to 100 to 200, validation accuracy is **60%**, 72%, 78%. What does the curve suggest, and what does it not guarantee?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Plot Plant Classifier train/validation performance at several **training** sizes and document the chosen next action.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Training sizes 20,40,80 yield validation accuracies 62%,**70%**,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Learning Curves, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

By the end, I can	How I will show it
Read training-size learning curves	Explain the deciding step, show a trace or test, and state one limit.
Describe variability across repeated development runs	Explain the deciding step, show a trace or test, and state one limit.
Avoid unsupported guarantees from a trend	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Learning Curves without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Read **training**-size **learning curve**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? More **training** plants improved the average in these supplied runs; **variability** remains.

1. State bounded pattern

2. Summarise larger size

3. Summarise first size

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State bounded pattern More **training** plants improved the average in these supplied runs; **variability** remains. Read the picture in order.

More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A30-plant experiment gives errors [4, 4, 4]. Which conclusion is supported by the supplied facts?

1. Mean remains4; this additional size does not improve the observed mean over20 plants.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean remains4; this additional size does not improve the observed mean over20 plants.

Learning curves are measured patterns, not a guarantee of improvement at every data size.

Repair a missing building block

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

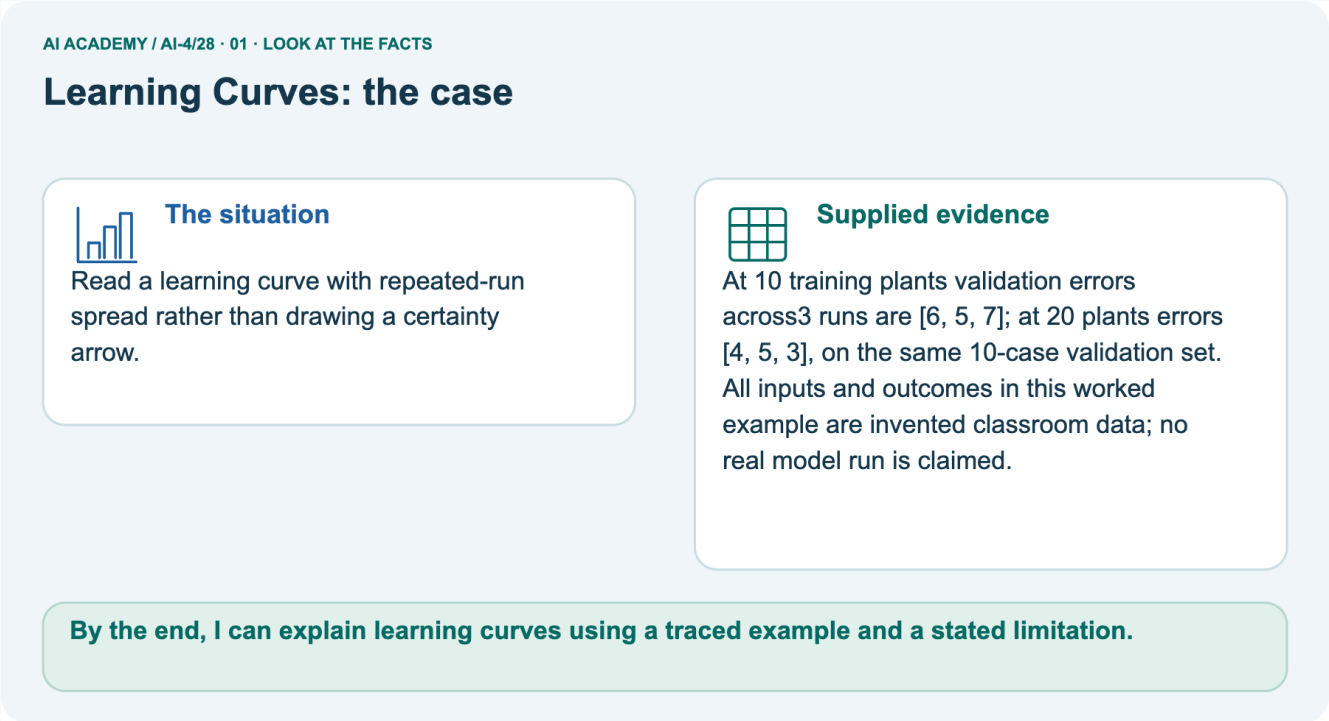
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Learning Curves graphic using its labels, deciding evidence and relevant limitation.

01. Learning Curves: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

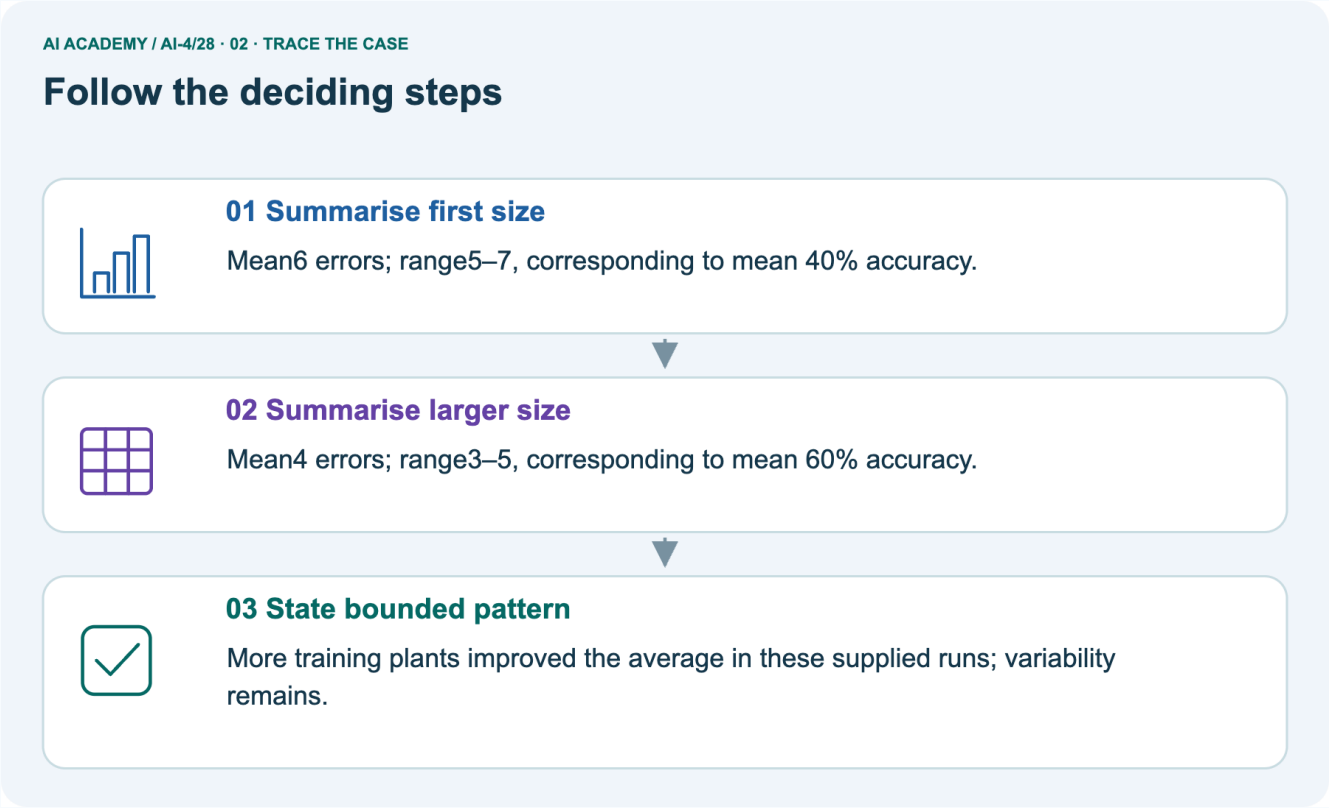
Read a **learning curve** with **repeat**ed-run spread rather than drawing a certainty arrow.

Supplied facts: At 10 **training** plants validation errors across3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Summarise first size: Mean6 errors; **range**5–7, corresponding to mean 40% accuracy. → Summarise larger size: Mean4 errors; **range**3–5, corresponding to mean **60%** accuracy. → State bounded pattern: More **training** plants improved the average in these supplied runs; **variability** remains.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Learning curves and variability at a glance

AI ACADEMY / AI-4/28 · 03 · INSPECT THE LESSON MODEL

Learning curves and variability at a glance

Part	Evidence or action	Interpretation
10 training rows	Validation 7 9 8	Mean 8 range 2
20 training rows	Validation 5 7 6	Mean 6 range 2
40 training rows	Validation 4 6 5	Mean 5 range 2

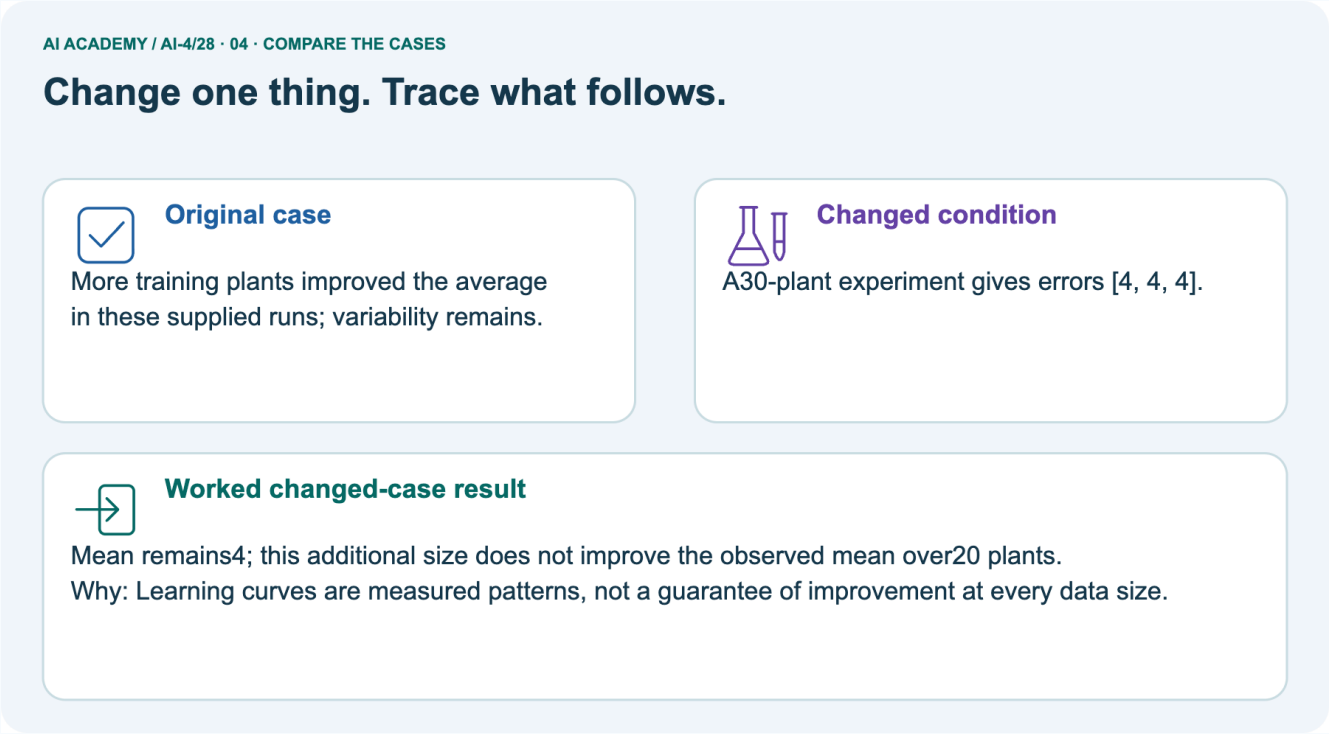
The decreasing gap is encouraging in this experiment. It does not establish that doubling again will halve error. A range from three repeats is a descriptive spread, not a confidence interval or a guarantee that future results fall inside it. Repeated runs can share records and therefore are not independent studies.

Follow the labelled picture with the matching explanation. Open the vector diagram

Learning curves and **variability** at a glance. 10 **training** rows | Validation 7 9 8 | Mean 8 **range** 2. 20 **training** rows | Validation 5 7 6 | Mean 6 **range** 2. 40 **training** rows | Validation 4 6 5 | Mean 5 **range** 2

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



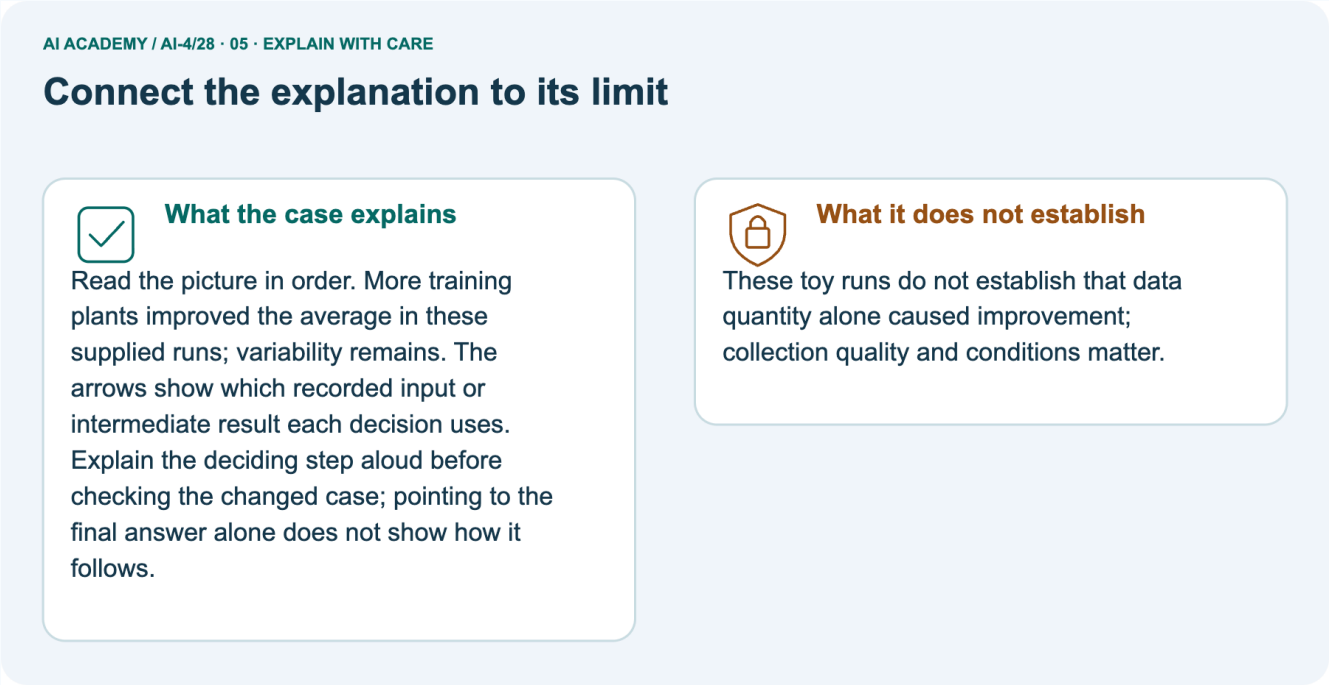
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: More **training** plants improved the average in these supplied runs; **variability** remains. Changed condition: A30-plant experiment gives errors [4, 4, 4].

Expected result: Mean remains4; this additional size does not improve the observed mean over20 plants. Reason: **Learning curve**s are measured patterns, not a guarantee of improvement at every data size.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

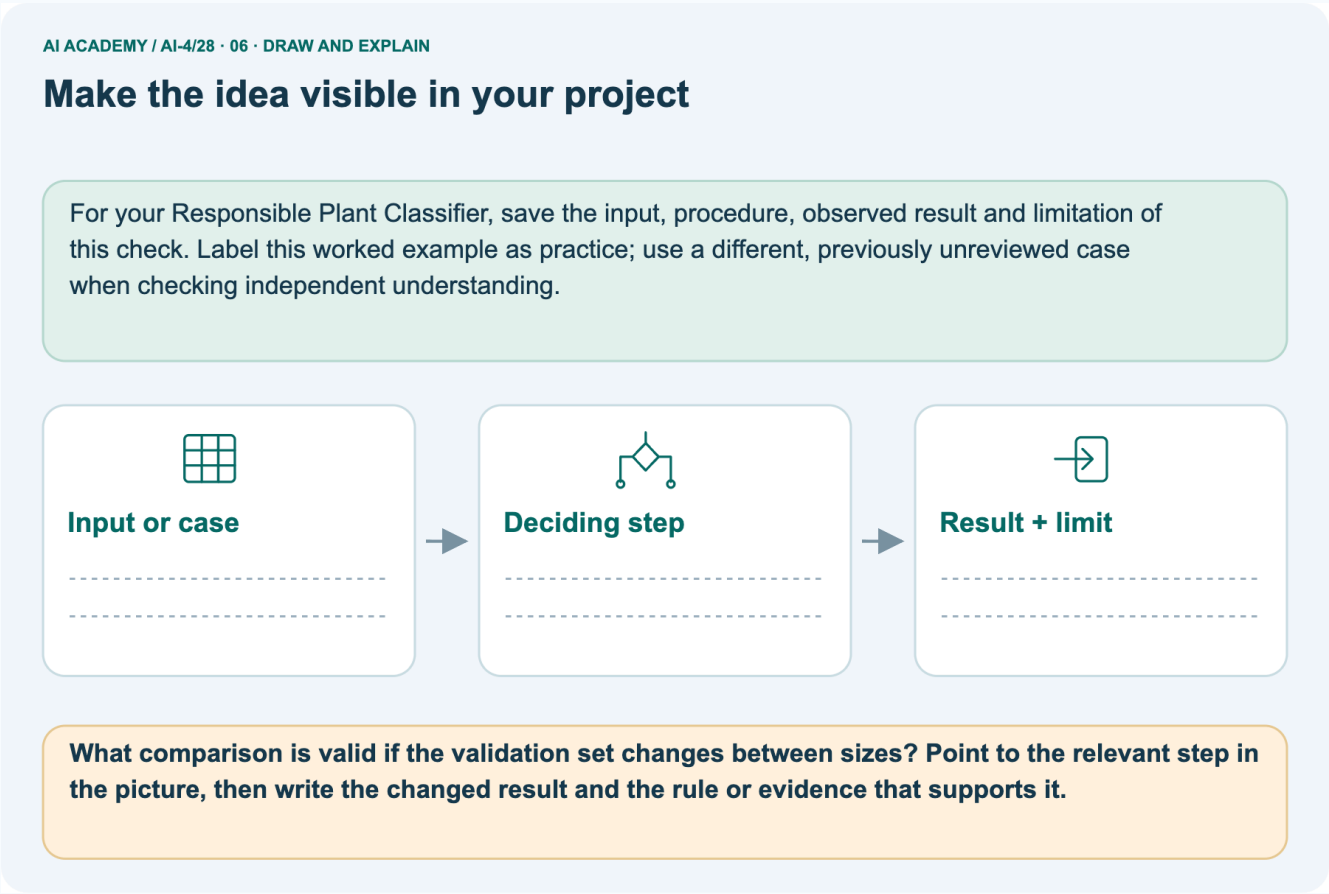
Read the picture in order. More **training** plants improved the average in these supplied runs; **variability** remains. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These toy runs do not establish that data quantity alone caused improvement; collection quality and conditions matter.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.